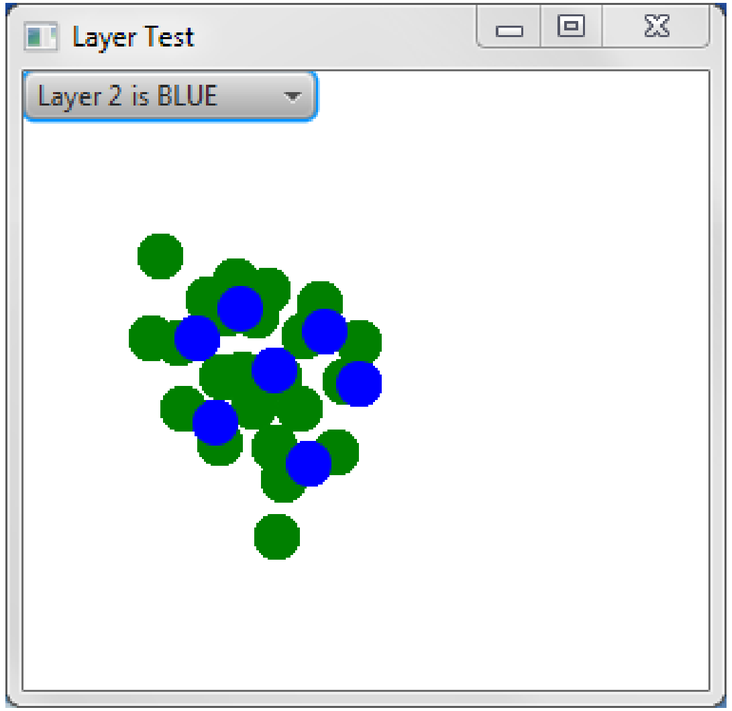
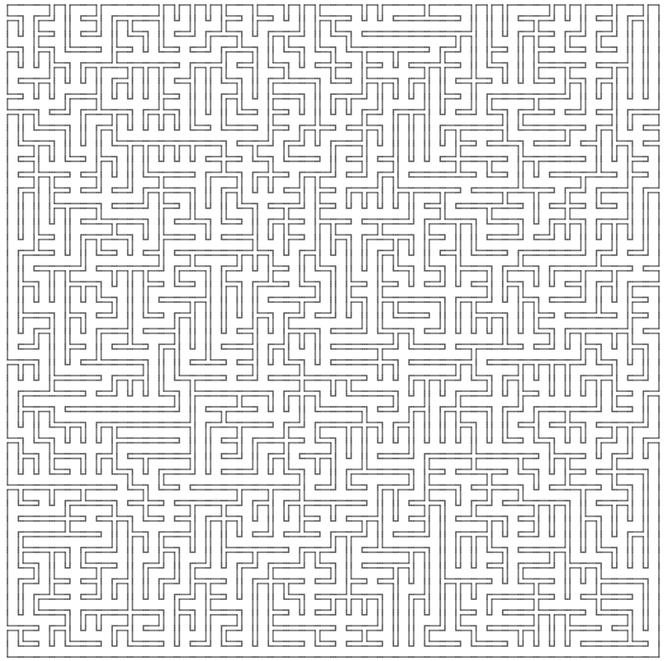
Canvas : Programmation de dessins 2D

Dany Siriphol – CODA 1 ère année - 2025

Canvas

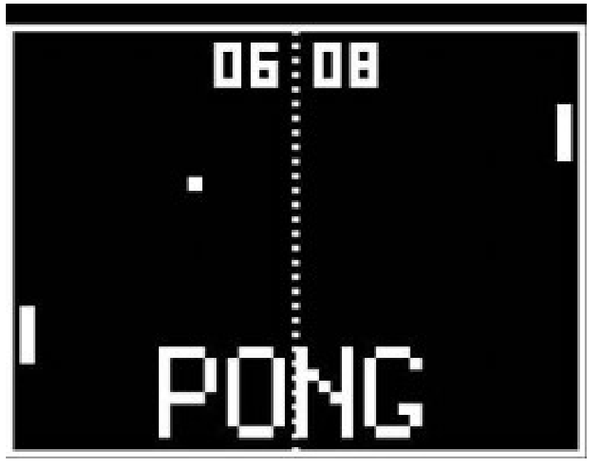
Le canvas est un composant JavaFX qui définît une zone dans laquelle on peut générer des dessins par programmation





Utilisation du canvas

- jeux 2D
- Graphiques personnalisés
- Animations
- dessin interactifs (ex : application type Paint)



Canvas

```
public void start(Stage stage) {

    Canvas canvas = new Canvas(300, 200);

    GraphicsContext gc = canvas.getGraphicsContext2D();
    BorderPane root = new BorderPane();
    root.setCenter(canvas);

    Scene scene = new Scene(root, 500, 500);
    stage.setTitle("Canvas");
    stage.setScene(scene);
    stage.show();
}
```

API pour les dessins 2D

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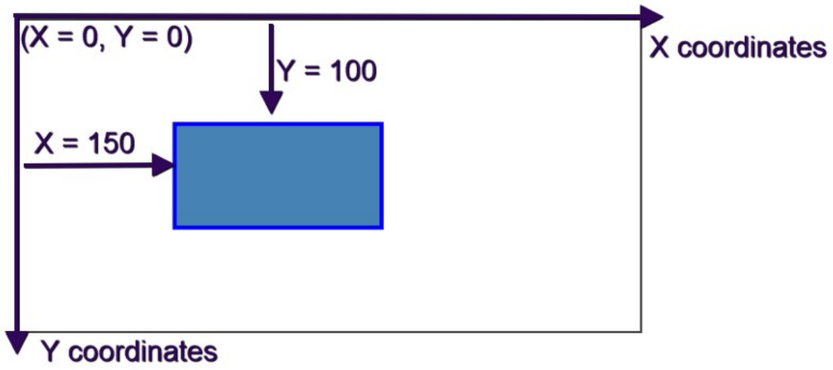
API de dessin

- Pour tracer des dessins, il faut tout d'abord charger le GraphicsContext du canvas

```
Canvas canvas = new Canvas(400, 300);
GraphicsContext gc = canvas.getGraphicsContext2D();
```

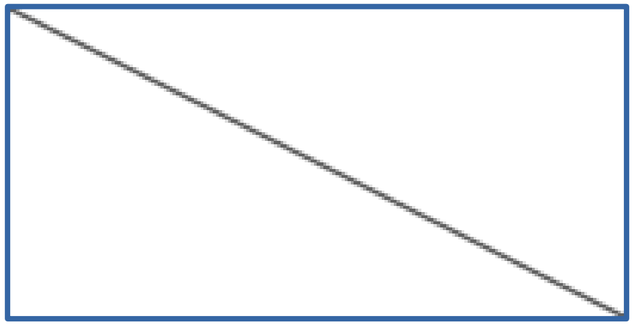
- Puis utiliser les méthodes de l'API pour tracer les formes souhaitées

```
gc.setFill(Color.BLACK);
gc.fillRect(50, 50, 200, 100);
```



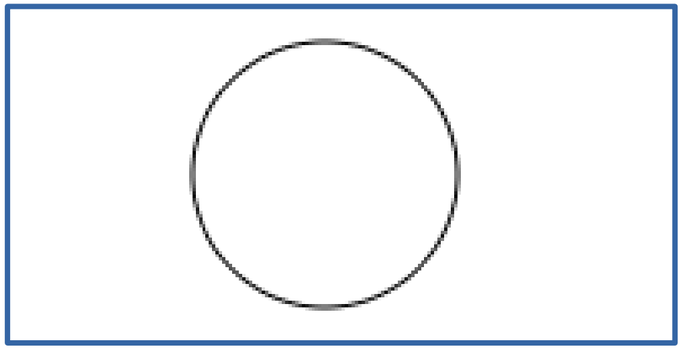
API de dessin

```
gc.strokeLine(0, 0, 400, 300);
```

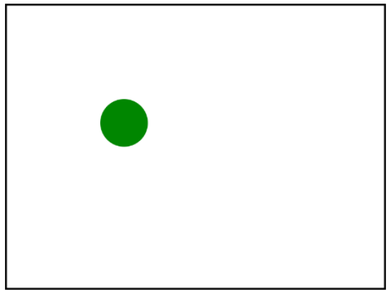


```
gc.moveTo(0, 0);
gc.lineTo(400, 300);
gc.stroke();
```

```
gc.strokeOval(100, 100, 100, 100);
```



```
gc.setFill(Color.GREEN);
gc.fillOval(100, 100, 50, 50);
```



API de dessin

Référence

https://docs.oracle.com/javase/8/javafx/api/javafx/scene/canvas/GraphicsContext.html

Style

```
gc.setFill(Color.RED);
gc.fillRect(50, 50, 100, 100);
```

```
gc.setStroke(Color.BLUE);
gc.setLineWidth(5);
gc.strokeLine(50, 50, 100, 100);
```

Style

```
// Rouge - Opacité 50%
gc.setFill(Color.rgb(255, 0, 0, 0.5));
gc.fillRect(50, 50, 100, 100);
```

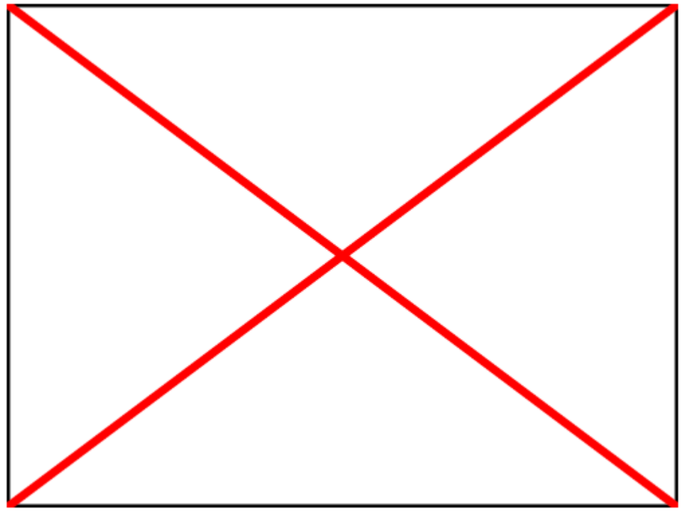
```
gc.setFill(new LinearGradient(0, 0, 1, 0, true,
        CycleMethod.NO_CYCLE,
        new Stop(0, Color.RED),
        new Stop(1, Color.BLUE))
);
gc.fillRect(50, 50, 100, 100);
```

Path

Un style est unique pour un seul tracé

```
gc.setStroke(Color.BLUE);
gc.setLineWidth(2);
gc.moveTo(0, 0);
gc.lineTo( 400, 300);

gc.setStroke(Color.RED);
gc.setLineWidth(5);
gc.moveTo(0, 300);
gc.lineTo(400, 0);
gc.stroke();
```



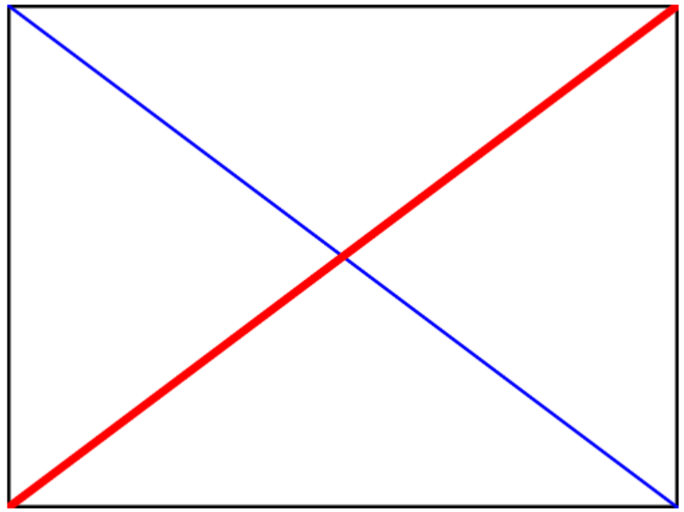
Path

Il est possible d’utiliser des Path pour créer plusieurs tracé et leur appliquer différents styles

```
gc.beginPath();

gc.setStroke(Color.BLUE);
gc.setLineWidth(2);
gc.strokeLine(0, 0, 400, 300);
gc.stroke();

gc.beginPath();
gc.setStroke(Color.RED);
gc.setLineWidth(5);
gc.strokeLine(0, 300, 400, 0);
gc.stroke();
```



Animation 2D

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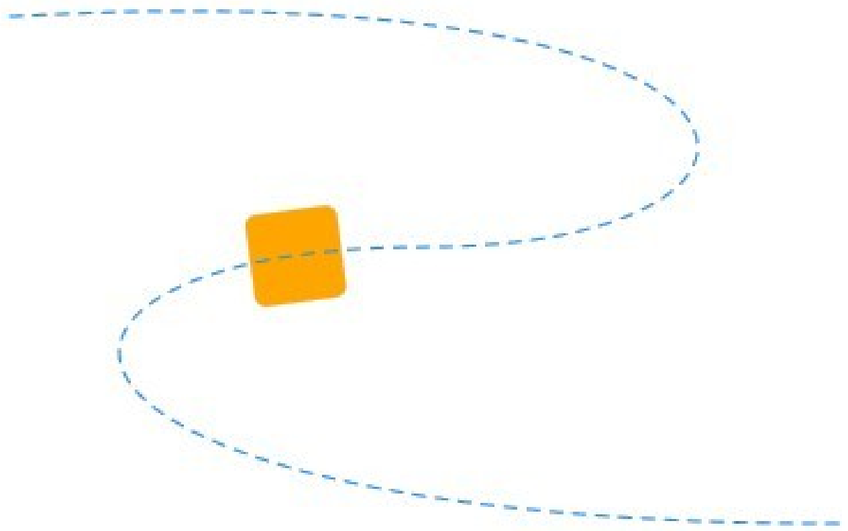
Principe

60 fois par seconde (60FPS) :

- Effacer le contenu du canvas
- Mettre à jour les données
- Redessiner le contenu



Démarrer une animation

AnimationTimer va appeler une fonction plusieurs fois par seconde en fonction de la fréquence de rafraîchissement de votre écran : 60 MHZ = 60 fois par seconde

```
AnimationTimer timer = new AnimationTimer() {
    @Override
    public void handle(long now) {

        System.out.println("hello");
    }
};
timer.start();
```

Exemple d’animation

```
private double x = 50;
private double speed = 2;
```

```
AnimationTimer timer = new AnimationTimer() {
    @Override
    public void handle(long now) {
        x += speed;
        gc.clearRect(0, 0, canvas.getWidth(), canvas.getHeight());
        gc.strokeOval(x , 50, 20, 20);
    }
};
timer.start();
```

Exemple d’utilisation

https://webdesign.tutsplus.com/21-ridiculously-impressive-html5-canvas-experiments--net-14210a

https://github.com/JoanClaret/html5-canvas-animation/blob/master/README.md

https://rosanko.wordpress.com/2014/01/17/21-ridiculously-impressive-html5-canvas-experiments/

Exercices